8

Homology modulo 2

The homology theory developed so far applies only to oriented simplicial complexes. In this chapter we shall show how to make adjustments in the theory so that it applies to unoriented simplicial complexes. Roughly speaking if an oriented simplex σ and the same simplex with opposite orientation, $-\sigma$, are to be regarded as giving the same chain, then $-1 = +1$ and so the coefficients must be regarded as lying in the group (or field) $\mathbb{Z}_2$ instead of in the group $\mathbb{Z}$. A satisfactory theory can be developed on these lines, though it is weaker than the theory with integral coefficients: for example the closed surfaces nT and $2nP$ have the same 'modulo 2' homology groups for any $n \geq 1$, and so cannot be distinguished by means of these groups.

The point of including this theory here is that with its aid certain results in Chapter 9 on the embedding of graphs in surfaces can be proved for both orientable and non-orientable closed surfaces.

Definitions 8.1. Let K be a simplicial complex of dimension n, and let $s_p^1, \ldots, s_p^{\alpha_p}$ be the p-simplexes of K for $p = 0, \ldots, n$. A *p-chain mod 2* on K is a formal sum

$$\lambda_1 s_p^1 + \cdots + \lambda_{\alpha_p} s_p^{\alpha_p}$$

where the coefficients λ_i are elements of the field $\mathbb{Z}_2$ so that each coefficient is 0 or 1. The set of p-chains mod 2 is denoted by $C_p(K; 2)$ (or $C_p(K; \mathbb{Z}_2)$) and it is a vector space over the field $\mathbb{Z}_2$ with the definitions of *addition* and *scalar multiplication* as follows:

$$\sum \lambda_i s_p^i + \sum \lambda_i' s_p^i = \sum (\lambda_i + \lambda_i') s_p^i$$
$$\lambda \left(\sum \lambda_i s_p^i \right) = \sum \lambda \lambda_i s_p^i, \quad \lambda \in \mathbb{Z}_2.$$

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For values of p other than $0, \ldots, n$ we define $C_p(K;2) = 0$.

The *boundary homomorphism mod 2*

$$\partial = \partial_p : C_p(K;2) \to C_{p-1}(K;2)$$

is defined for $0 < p \leqslant n$ by

$$\partial\left(\sum \lambda_i s_p^i\right) = \sum \lambda_i \partial s_p^i,$$

where

$$\partial(v^0 \ldots v^p) = \sum_{i=0}^{p} (v^0 \ldots \widehat{v^i} \ldots v^p),$$

and is defined to be trivial otherwise. In fact ∂ is a linear map.

The *augmentation mod 2* is the linear map

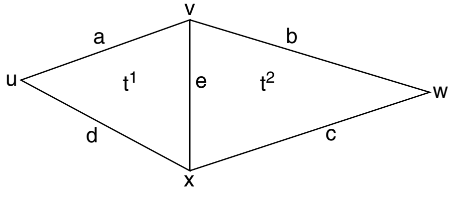
$$\varepsilon : C_0(K;2) \to \mathbb{Z}_2$$

defined by

$$\varepsilon\left(\sum \lambda_i s_0^i\right) = \sum \lambda_i.$$

Thus, in the diagram below, we have (all mod 2)

$$\begin{aligned}
\partial t^1 &= a + e + d \\
\partial t^2 &= b + c + e \\
\partial(t^1 + t^2) &= a + b + c + d + 2e \\
&= a + b + c + d \quad \text{since } 2e = 0 \\
\partial a &= u + v, \quad \text{etc.} \\
\partial(a + b + c + d) &= u + v + v + w + w + x + x + u \\
&= 0.
\end{aligned}$$



As another example, consider the diagram of a Möbius band in 4.15. Removing all arrows which indicate orientations we have

$$\partial(t^1 + t^2 + t^3 + t^4 + t^5) = e^2 + e^3 + e^4 + e^5 + e^6 \pmod 2;$$

that is 'the mod 2 boundary of the whole band is its rim'. Putting it the other way round, the rim is, mod 2, a boundary. This is false in ordinary homology: there is no 2-chain on the oriented K whose boundary is $e^2 + \cdots + e^6 \in C_1(K)$. This suggests that, for simple closed polygons, the property of enclosing a region is closely connected with being a boundary mod 2. (This will be made precise and proved in 9.19.)

An example where the Möbius band occurs as a region in something bigger is given by the Klein bottle of 2.2(4), where the shaded triangles form a Möbius band. In this case the rim of the band represents, in ordinary homology, $\pm 2\{a + b + c\}$, which is not zero (compare the last sentence of 6.14(1)). When we pass to homology mod 2 the corresponding class is zero, since the rim is, mod 2, a boundary.

Remarks 8.2. ★(1) We can form the group of p-chains with coefficients in any abelian group A in an analogous way to that given above for $A = \mathbb{Z}_2$, and this group is denoted by $C_p(K; A)$. The case $A = \mathbb{Z}$ is the one used hitherto. In the formula for ∂, a sign $\pm$ must be introduced, $+$ if i is even and – if i is odd, in order that $\partial \circ \partial$ should be zero, but for $A = \mathbb{Z}_2$ the sign is redundant. If A is a field, such as the field of real numbers, rational numbers or complex numbers, then $C_p(K; A)$ is a vector space over A, using the same definitions of scalar multiplication and addition as for $A = \mathbb{Z}_2$; furthermore in that case ∂ is a linear map. ★

(2) Suppose that $f : A \to B$ is a map between vector spaces over $\mathbb{Z}_2$, and that $f(a_1 + a_2) = f(a_1) + f(a_2)$ for all a_1, a_2, in A. Then f is automatically a linear map, i.e. f also satisfies $f(\lambda a) = \lambda f(a)$ for all $\lambda \in \mathbb{Z}_2$ and $a \in A$. (There are only two cases, $\lambda = 0$ and $\lambda = 1$, to check!)

Chain complexes mod 2, augmented chain complexes mod 2, boundaries and cycles are all defined as in the case of $\mathbb{Z}$ coefficients (4.5 and 4.8). We also have the following result.

Proposition 8.3. *For any p,*

$$\partial_{p-1} \circ \partial_p : C_p(K; 2) \to C_{p-2}(K; 2)$$

is the trivial homomorphism. Furthermore

$$\varepsilon \circ \partial_1 : C_1(K;2) \to \mathbb{Z}_2$$

is trivial. □

Thus $\operatorname{Im} \partial_{p+1} \subset \operatorname{Ker} \partial_p$ for all p, and we can at any rate form a quotient group

$$H_p(K;2) = \operatorname{Ker} \partial_p / \operatorname{Im} \partial_{p+1} = Z_p(K;2)/B_p(K;2)$$

where $Z_p(K;2)$ and $B_p(K;2)$ are regarded as just abelian groups. Then $H_p(K;2)$ is called the pth *homology group of* K mod 2 (or *with* $\mathbb{Z}_2$ *coefficients*). In fact $H_p(K;2)$ is a vector space over $\mathbb{Z}_2$ with scalar multiplication defined by

$$\lambda(z + B_p(K;2)) = \lambda z + B_p(K;2)$$

where $z \in Z_p(K;2)$ and $\lambda \in \mathbb{Z}_2$. Thus as a vector space, and hence as an abelian group also,

$$H_p(K;2) \cong \mathbb{Z}_2 \oplus \mathbb{Z}_2 \oplus \cdots \oplus \mathbb{Z}_2.$$

The number of summands is the dimension of the vector space $H_p(K;2)$, and is called the pth *connectivity number* of K. It is denoted $\widehat{\beta}_p(K)$.

There is no analogue of 'torsion coefficients', for every vector space is free in the sense that

$$\lambda x = 0 \quad (\lambda \in \text{field}, x \in \text{vector space}) \Rightarrow \lambda = 0 \quad \text{or} \quad x = 0.$$

Of course considered just as an abelian group, $\mathbb{Z}_2 \oplus \cdots \oplus \mathbb{Z}_2$ is far from free, since $\lambda x = 0$ for any $x \in \mathbb{Z}_2 \oplus \cdots \oplus \mathbb{Z}_2$ when $\lambda = 2 \in \mathbb{Z}$.

Two elements z, z' in $Z_p(K;2)$ are called *homologous mod 2* if $z - z' \in B_p(K;2)$. We then write $z \sim z'$ (mod 2). The homology class mod 2 of $z \in Z_p(K;2)$, namely $z + B_p(K;2)$, is denoted $\{z\}$ or $\{z\}_K$.

We also define the *reduced group* (again a vector space)

$$\widetilde{H}_0(K;2) = \operatorname{Ker} \varepsilon / \operatorname{Im} \partial_1.$$

Examples 8.4. (1) Let M be a closed surface and let $c \in C_2(M;2)$ be the sum of all the triangles of M. Then c is a cycle, because every edge of M is on exactly two triangles of M. Notice that this does not assume M to be orientable. Using a similar argument to that in 4.18 it follows that $H_2(M;2) \cong \mathbb{Z}_2$, with c (strictly $\{c\}$) as a generator. We call c the *fundamental cycle mod 2* on M. Thus

$H_2(M;2)$ does not distinguish between orientable and non-orientable closed surfaces. (But see 8.9(3).)

(2X) Suppose that K is connected. A basis for $Z_1(K;2)$ is obtained in the following way, exactly analogous to that for $Z_1(K)$ in 4.9(2). Let T be a maximal tree in the 1-skeleton K^1 of K, and let $e^1, \ldots, e^\mu$ be the edges of K^1 not in T. Then for each $i = 1, \ldots, \mu$, $T + e^i$ contains a unique loop and the sum of the edges in this loop is a 1-cycle z^i mod 2. We have: $z^1, \ldots, z^\mu$ is a basis for $Z_1(K;2)$. If K is not connected put together bases for the 1-cycles in each component of K.

In particular if K is a connected graph ($K^1 = K$) then $H_1(K;2) \cong \mathbb{Z}_2 \oplus \cdots \oplus \mathbb{Z}_2$ (μ summands) with basis $z^1, \ldots, z^\mu$. If K has several components then $H_1(K;2)$ is the direct sum of the $H_1(K_i;2)$ where K_i runs through the components of K.

(3X) Let K be obtained from a closed surface M by removing a single 2-simplex s, and let z be the 1-cycle mod 2 on K given by the sum of the edges of s. Then $z \sim 0$ (mod 2) on K whether or not M is orientable – compare 7.6(3).

(4) $H_0(K;2)$ has dimension equal to the number of components of K, with basis consisting of the homology classes mod 2 of one vertex from each component. The proof given in 4.11 of the corresponding statement about $H_0(K)$ also proves the present statement, subject only to the obvious changes from $\mathbb{Z}$ to $\mathbb{Z}_2$ in various places. Similarly, if K is non-empty,

$$H_0(K;2) \cong \widetilde{H}_0(K;2) \oplus \mathbb{Z}_2.$$

Again the proof given before (4.13) is valid, remembering that short exact sequences of vector spaces and linear maps always split (see after A.19).

(5X) Let K be the triangulation of a Möbius band given in 4.15. The calculation of $H_1(K;2)$ proceeds in exactly the same way as that for $H_1(K)$, except that all arrows can be removed and minus signs changed to plus signs. We can either regard the presentations as defining abelian groups in which, additionally to the relations given, every element has order two, or better, as defining vector spaces over $\mathbb{Z}_2$ as quotients of vector spaces by subspaces. In the latter case the theory is exactly parallel to that expounded in the Appendix and we can manipulate the presentations until a basis for the quotient vector space emerges. Of course these manipulations are enormously simplified by the fact that $1 + 1 = 0$, and at the end of the calculation there will be generators but no relations in the presentation.

The result for the Möbius band is

$$H_1(K;2) \cong \mathbb{Z}_2 \quad \text{with basis } \{z^1\}.$$

Note that $\{e^2 + e^3 + e^4 + e^5 + e^6\} = 2\{z^1\} = 0 \pmod 2$ (see the discussion before 8.2).

The definitions of *relative homology groups mod* 2 and *relative boundary homomorphisms mod* 2 are exactly parallel to those in Chapter 4, integer coefficients being replaced by coefficients in $\mathbb{Z}_2$. These relative groups are also vector spaces over $\mathbb{Z}_2$ and the homomorphisms are linear maps (see 8.2(2)). The statement and proof of 4.24 are still valid: given $c \in C_p(K, L; 2)$ then $c \in Z_p(K, L; 2)$ if and only if $\partial c \in C_{p-1}(L; 2)$; and $c \in B_p(K, L; 2)$ if and only if $\partial c' - c \in C_p(L; 2)$ for some $c' \in C_{p+1}(K; 2)$.

The homomorphisms i_*, j_* and ∂_* defined in 4.30, 4.32 and 4.34 carry over to the mod 2 situation and the *mod 2 homology sequence of a pair* (K, L) is exact. The statement and proof in 6.1 apply to the present case merely by changing integers to integers mod 2 throughout. The *reduced* sequence is also exact provided $L \neq \emptyset$.

The *excision theorem* is also true in mod 2 homology and the proof given in 6.4 carries over. The same is true of *collapsing* (6.6): if $K \searrow L$ then $i_* : H_p(L; 2) \to H_p(K; 2)$ is an isomorphism for all p. It follows from the proof of 6.8 that the mod 2 homology groups of any cone CK are $H_p(CK; 2) = 0$, $p > 0$; $H_0(CK; 2) \cong \mathbb{Z}_2$.

All the invariance properties of Chapter 5 carry over to the mod 2 case.

The calculation of $H_1(M; 2)$ for a closed surface M is simpler than that of $H_1(M)$ since the analogue of 6.12 is:

Let K be obtained from M by removing a single 2-simplex. Then $i_* : H_1(K; 2) \to H_1(M; 2)$ is an isomorphism.

The proof is provided by the proof of 6.12(1), which carries over whether M is orientable or not. Hence we have the following result (compare 6.13).

Theorem 8.5. *Let M be a closed surface. Then $H_1(M; 2)$ is as follows:*

(1) $H_1(M; 2) = 0$ *when M has standard form aa^1.*

(2) $H_1(M; 2) \cong (\mathbb{Z}_2)^k$ *when M has standard form $a_1 a_1 \ldots a_k a_k$. A basis is $\{a_1\}, \ldots, \{a_k\}$.*

(3) $H_1(M; 2) \cong (\mathbb{Z}_2)^{2h}$ *when M has standard form $a_1 b_1 a_1^{-1} b_1^{-1} \ldots a_h b_h a_h^{-1} b_h^{-1}$. A basis of $\{a_1\}, \{b_1\}, \ldots, \{a_h\}, \{b_h\}$. (Cycles representing this basis are obtained by removing the arrows from those drawn in 6.14(2).)* □

Notice that $H_1(2nP; 2) \cong H_1(nT; 2)$ for any integer $n \geqslant 1$. Thus $2nP$ and nT have the same homology groups, mod 2, for any $n \geqslant 1$ (see 8.4(1)). Not surprisingly, the mod 2 groups are a weaker tool than the groups with integer coefficients.

The theorem corresponding to the Euler characteristic formula of 6.16 is the following. Let K be a simplicial complex of dimension n. Recall that $\widehat{\beta}_p = \dim H_p(K;2)$ and that $\alpha_p =$ number of p-simplexes of $K = \dim C_p(K;2)$.

Theorem 8.6.

$$\chi(K) = \sum_{p=0}^{n}(-1)^p \alpha_p = \sum_{p=0}^{n}(-1)^p \widehat{\beta}_p.$$

The proof is simpler than that of 6.16 since it depends only on the 'rank plus nullity' theorem of linear algebra rather than on the more difficult theorem A.31. (The rank plus nullity theorem states that if $f : V \to W$ is a linear map between vector spaces, and V is finite dimensional, then $\dim(\mathrm{Im} f) + \dim(\mathrm{Ker} f) = \dim V$.) □

8.7. Alternating sum of dimensions theorem. *Suppose that*

$$0 \to A_n \to A_{n-1} \to \cdots \to A_1 \to A_0 \to 0$$

is an exact sequence of vector spaces and linear maps. Then

$\sum_{i=0}^{n}(-1)^i \dim A_i = 0.$ *(Compare A.32(3).)* □

Examples 8.8. (1X) Let K and L be as in 4.29(3) (Möbius band and its rim). Then $H_1(K, L; 2) \cong \mathbb{Z}_2$ with generator $\{e^1\}$ and $H_2(K, L; 2) \cong \mathbb{Z}_2$ with generator $\{t^1 + t^2 + t^3 + t^4 + t^5\}$. These follow either from a direct calculation as in 4.29(3) or from the mod 2 homology sequence of (K, L) and 8.4(5) above (the mod 2 homology groups of K). Compare 6.3(2), which does the calculations with ordinary homology.

(2X) Suppose that L is a connected subcomplex of K. Show that $j(Z_1(K;2)) = Z_1(K, L; 2)$. Is this ever true if L is not connected? (Compare 4.29(4).)

(3X) Repeat Example 6.5 (removal of a 2-simplex from a two-dimensional simplicial complex) with homology mod 2.

(4X) Let S^n denote the set of proper faces of an $(n+1)$-simplex ($n \geqslant 1$). Then $H_p(S^n; 2) = 0$ ($p \neq 0, n$); $H_0(S^n; 2) \cong H_n(S^n; 2) \cong \mathbb{Z}_2$. (Compare 6.11(4).)

The statement and proof (7.1, 7.2) of the exactness of the *Mayer–Vietoris sequence* of a triad $(K; L_1, L_2)$ go over readily to the mod 2 case. Of course there is no longer any need for the minus sign in the definition of ϕ. All the examples of 7.4 work equally well with mod 2 homology groups (where $\mathbb{Z}$ must be changed to $\mathbb{Z}_2$ in (4)), and the reader is invited to try them. Likewise the *homology sequence of a triple* (M, N, K) is exact in mod 2 theory. (See 7.7, 7.8.)

Examples 8.9. (1X) Let M_1 and M_2 be closed surfaces and let K_1 and K_2 be obtained from them by removing a 2-simplex from each. Use the Mayer–Vietoris sequence (mod 2) to show that

$$i_* : H_1(K_\alpha; 2) \to H_1(M_\alpha; 2)$$

is an isomorphism for $\alpha = 1, 2$. Hence use the sequence (mod 2) to show that

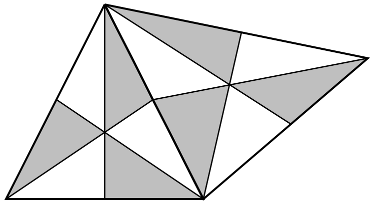
$$H_1(M_1 \# M_2; 2) \cong H_1(M_1; 2) \oplus H_1(M_2; 2).$$

Compare 7.6(2)–(3).

(2) In the calculation, given in 7.5, of the first homology groups of triangulations of torus and Klein bottle by sticking together two cylinders, the only difference between the two cases is a difference of sign, and this disappears when we pass to mod 2 homology groups. Thus, as noted in 8.5 above, these two closed surfaces have isomorphic first homology groups.

(3X) It might be supposed that there was no way of distinguishing between an orientable and a non-orientable closed surface using exclusively mod 2 homology. This is not so. Let M be a closed surface and let M' be the first barycentric subdivision of M. Clearly M' is also a closed surface equivalent to M and M' is orientable if and only if M is orientable. Let w_p $(p = 0, 1, 2)$ be the element of $C_p(M'; 2)$ given by the sum of all the p-simplexes of M'. It is straightforward to verify that w_1 is a 1-cycle mod 2 – this just uses the fact that every vertex of M' is an end-point of an even number of edges. In fact

$$\{w_1\} = 0 \Leftrightarrow M' \text{ (and hence } M\text{) is orientable.}$$



The reader may like to find an informal proof of this. The diagram is intended to be a hint.

Notice that w_0 and w_2 are also cycles. The elements $\{w_p\} \in H_p(M'; 2), p = 0, 1, 2$ are called the *Whitney* (or *Stiefel–Whitney*) *homology classes of* M. They are an example of 'characteristic classes' which are of great importance in algebraic topology. (See for example the books of Spanier and Milnor & Stasheff listed in the References. A modern reference for the homology classes

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mentioned above is Halperin & Toledo (1972). This article gives a proof of Whitney's original theorem, for which he did not publish a proof himself. There are many subsequent developments, for example Goldstein & Turner (1976) dispense with the need for taking a barycentric subdivision.)

★(4) If the ordinary homology groups $H_p(K)$ of a simplicial complex K are known, then it is actually possible to derive the groups $H_p(K;2)$. The formula is

$$H_p(K;2) \cong (H_p(K) \otimes \mathbb{Z}_2) \oplus (\mathrm{Tor}(H_{p-1}(K), \mathbb{Z}_2)).$$

We shall not go into the details of the tensor product $\otimes$ and the torsion product Tor here, but the answer can be worked out for any given K using the following rules. (See for example the book of Hilton & Wylie listed in the References; note that they use $*$ for Tor. The formula is known as the *Universal Coefficients Theorem* for homology. See also the book *Elements of Homology Theory* by Prasolov.)

(i) Direct sums can be 'multiplied out', e.g.

$$\begin{aligned}&(A_1 \oplus A_2) \otimes (B_1 \oplus B_2)\\ &\quad \cong (A_1 \otimes B_1) \oplus (A_1 \otimes B_2) \oplus (A_2 \otimes B_1) \oplus (A_2 \otimes B_2),\end{aligned}$$

$$\begin{aligned}&\mathrm{Tor}(A_1 \oplus A_2, B_1 \oplus B_2)\\ &\quad \cong \mathrm{Tor}(A_1, B_1) \oplus \mathrm{Tor}(A_1, B_2) \oplus \mathrm{Tor}(A_2, B_1) \oplus \mathrm{Tor}(A_2, B_2);\end{aligned}$$

(ii) $A \otimes B \cong B \otimes A$; $\mathrm{Tor}(A, B) \cong \mathrm{Tor}(B, A)$;

(iii) $\mathbb{Z} \otimes \mathbb{Z}_k \cong \mathbb{Z}_k$; $\mathbb{Z} \otimes \mathbb{Z} \cong \mathbb{Z}$; $A \otimes 0 = 0$;
$\mathbb{Z}_m \otimes \mathbb{Z}_n \cong \mathbb{Z}_{(m,n)}$ where (m, n) = greatest common divisor of m and n;

(iv) $\mathrm{Tor}(\mathbb{Z}, \mathbb{Z}_k) = 0$; $\mathrm{Tor}(\mathbb{Z}, \mathbb{Z}) = 0$; $\mathrm{Tor}(A, 0) = 0$; $\mathrm{Tor}(\mathbb{Z}_m, \mathbb{Z}_n) \cong \mathbb{Z}_{(m,n)}$.

Thus for example take $K = P$, the projective plane, so that $H_0(P) \cong \mathbb{Z}$, $H_1(P) \cong \mathbb{Z}_2$, $H_2(P) = 0$. Then,

$$\begin{aligned}H_0(P;2) &\cong (\mathbb{Z} \otimes \mathbb{Z}_2) \oplus \mathrm{Tor}(0, \mathbb{Z}_2) \cong \mathbb{Z}_2\\ H_1(P;2) &\cong (\mathbb{Z}_2 \otimes \mathbb{Z}_2) \oplus \mathrm{Tor}(\mathbb{Z}, \mathbb{Z}_2) \cong \mathbb{Z}_2\\ H_2(P;2) &\cong (0 \otimes \mathbb{Z}_2) \oplus \mathrm{Tor}(\mathbb{Z}_2, \mathbb{Z}_2) \cong \mathbb{Z}_2. \bigstar\end{aligned}$$

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